

```
1  library ieee;
2  use ieee.std_logic_1164.all;
3
4  entity two_to_fourDecoder is
5      port(input2_4De: in std_logic_vector(1 downto 0);
6            en:        in std_logic;
7            output2_4De: out std_logic_vector(3 downto 0));
8  end;
9
10 architecture synth of two_to_fourDecoder is
11 begin
12     process (en) begin
13         if en = '1' then
14             output2_4De(3) <= input2_4De(1) and input2_4De(0);
15             output2_4De(2) <= input2_4De(1) and not input2_4De(0);
16             output2_4De(1) <= not input2_4De(1) and input2_4De(0);
17             output2_4De(0) <= not input2_4De(1) and not input2_4De(0);
18         else
19             output2_4De(3) <= '0';
20             output2_4De(2) <= '0';
21             output2_4De(1) <= '0';
22             output2_4De(0) <= '0';
23         end if;
24     end process;
25 end;
```